

HUMAN INFLUENCES ON ECOSYSTEMS

human influences on ecosystems

Humans have increased food production through ::

- agricultural machinery
 - ↳ to use larger areas of land and improve efficiency
- chemical fertilisers
 - ↳ to improve yields
- insecticides
 - ↳ to improve quality and yield
- herbicides
 - ↳ to reduce competition with weeds
- selective breeding
 - ↳ to improve production by crop plants and livestock

Monoculture ::

monoculture farming means that on a given area of agricultural land, only one type of crop is grown. It doesn't happen naturally

- large scale monoculture of crop plants leads to =

- ↳ reduced diversity of ecosystem
- ↳ increased pests
- ↳ problems associated with insecticides

Intensive livestock production

Leads to =

- ↳ diseases spreading easily
- ↳ welfare issues between livestock

Biodiversity ::

the number of different species that live in an area

Reasons for habitat destruction ::

- ↳ increased area for housing, crop plant and livestock production
- ↳ extraction of natural resources
- ↳ fresh water and marine pollution

* through altering food webs and food chains, humans can have a negative impact on habitats.

Deforestation ::

- the clearing of trees (usually on a large scale)

Undesirable effects of deforestation ::

- ↳ reduced biodiversity
- ↳ extinction of species
- ↳ loss of soil
- ↳ increase of CO_2 in atmosphere.

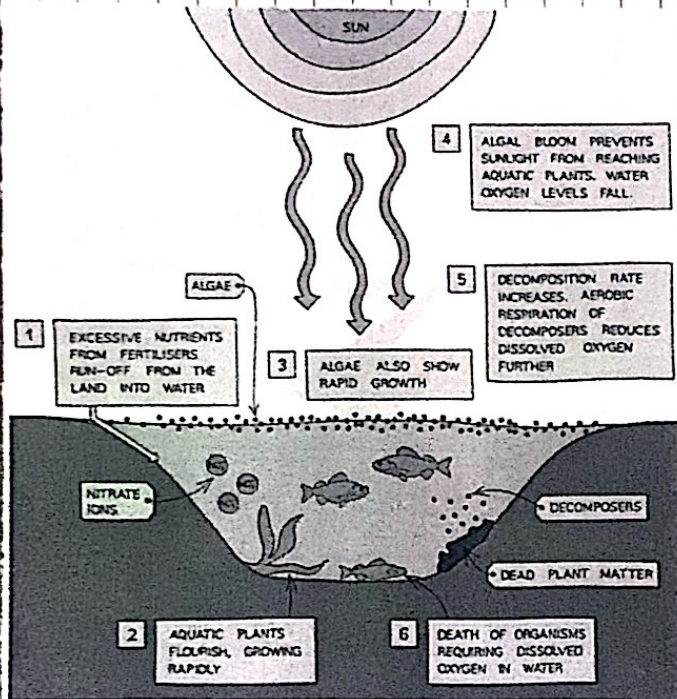
Untreated sewage::

- provides a good source of food for bacteria which increase rapidly
 - ↳ depleting the oxygen dissolved in the water (as they respire aerobically)
 - ↳ causing the death of aquatic organisms such as fish.

Excess fertilisers::

- causes algal bloom which then die and provide a good source of food for decomposing bacteria which increase rapidly
 - ↳ depleting the oxygen dissolved in the water (as they respire aerobically)
 - ↳ causing death of aquatic organisms such as fish

Eutrophication::



Non-biodegradable plastics::

in marine habitats =

- animals often try to eat plastic or become caught in it, leading to injuries and death
- as the plastic breaks down, it releases toxins that affect the organisms
- once broken down into very small particles, it is ingested by animals and enters the food chain

on land =

- plastic is generally disposed of by burying in landfills
- as it breaks down, it releases toxins into the surrounding soil and as such the land is no good for growing crops or grazing animals

Methane and Carbon dioxide::

- both gases insulate the Earth and act as a 'blanket' around the atmosphere
- higher levels of both have led to global warming and climate change
- enhanced greenhouse effect is where there is an increase in greenhouse gases thus more heat absorbed by the atmosphere

Sources

- * burning fossil fuels increases CO_2
- * keeping livestock generates methane gas
- * global warming melts the permafrost in sub-polar regions, which results in even more trapped methane being released in the atmosphere

Sustainable resource::

- a resource which is produced as rapidly as it is removed from the environment so that it doesn't run out
- * Some resources can be conserved and managed sustainably such as forests and fish stocks
 - ↳ forests can be conserved through education, protected areas, quotas, and replanting
 - ↳ fish stock can be conserved through education, closed seasons, protected areas, controlled net types/mesh size, quotas & monitoring

Endangerment & Extinction::

- caused by climate change, habitat destruction, hunting, over harvesting, pollution and introduced species.
 - ↳ endangered species are conserved through monitoring/protecting species and habitats, education, captive breeding programmes, and seed banks

Reasons for conservation programmes ::

- maintaining/increasing biodiversity
- reducing extinction
- protecting vulnerable ecosystems
- maintaining ecosystem functions - nutrient cycling, resource provision (food, drugs, fuel and genes)

AI and IVF::

artificial insemination (AI)

- allows large number of offspring to be produced w/o the need for conventional sexual intercourse between males and females
- in vitro fertilisation (IVF)
 - this allows gametes with known alleles to be used in ensuring the next generation remains biodiverse

- * if its population decreases, a species will experience reduced genetic variation

- ↳ this renders the species more at risk to environmental change
 - ↳ the species is less resilient and has a greater risk of extinction